



A few definitions just in case:

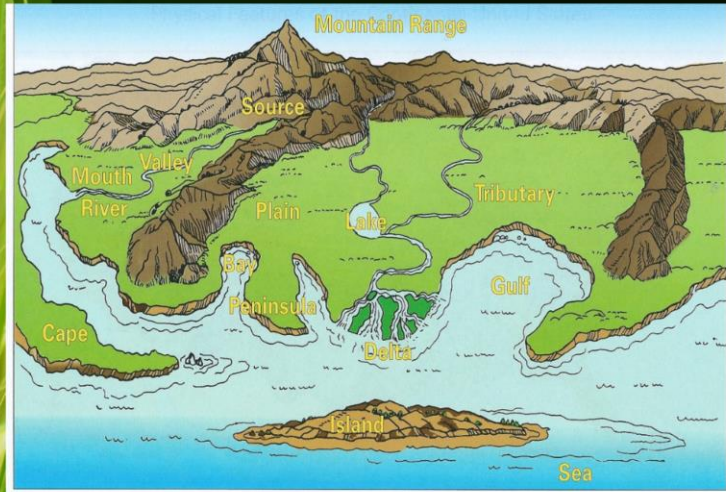
Fertilizer: a chemical or natural product used to add nutrients to the soil

Pesticide: a chemical or natural product used to prevent damage to crops by insects

Cultivate: grow

Domesticate: raise for human use

Physical Geographic Patterns



Every continent has a variety of landforms. Some landforms are more suitable for farming/agriculture than others and some are more suited to one type of agriculture compared to others.

Mountains – elevated and rugged topography makes farming difficult, however, mountains have bands of climate zones at different elevations. For example, it may be hot and dry at the base of the mountain, but halfway up it will be cooler with more precipitation, and the top of the mountain is freezing and has an ice cap.

Plains - a plain is a flat, sweeping landmass that generally does not change much in elevation. Plains occur as lowlands along valleys or at the base of mountains, as coastal plains, and as plateaus (elevated plains).

River Valleys - a low area between hills or mountains typically with a river running through it. The land in the valley is very fertile because when the river floods it carries large amounts of organic sediment over the flood plain (the area that floods). As the water recedes, the organic matter is left behind like a natural fertilizer. The earliest ancient civilizations developed in river valleys.



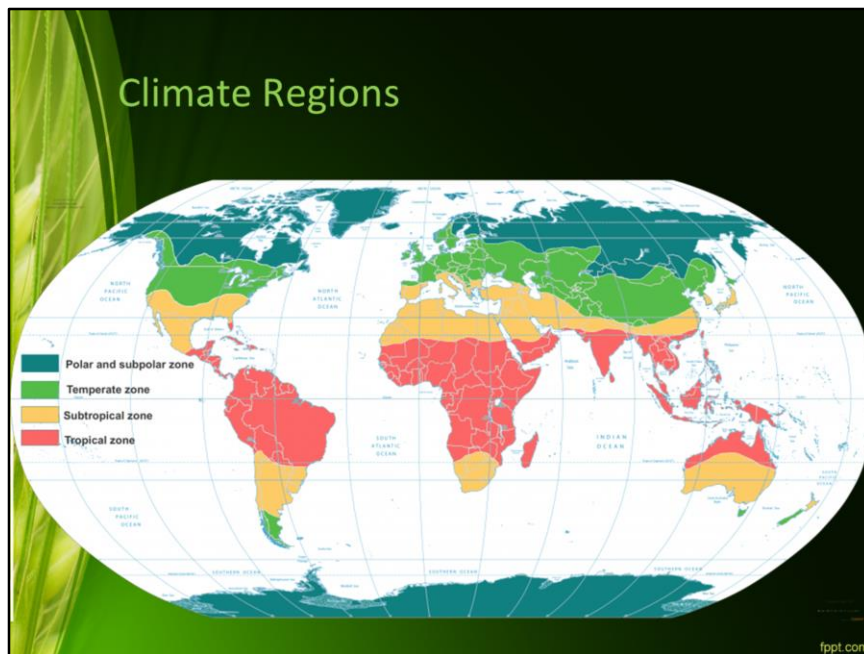
The US covers much of the North American continent, so it has a wide variety of landforms.

The light green areas above are low-lying coastal plains and river valleys. Notice how wide the plains are on the east coast and how narrow they are on the west coast. This land was forested in its natural state. Large swatches of trees were cleared for agriculture in the early days of the US.

The yellow area is the Great Plains region (which actually continues north into Canada to the Grand Prairies region). The Great Plains are flat, with coastal plains in south Texas that give way to plateaus in the middle part of the state (plateau is an elevated plain – coastal plains are close to sea level, while plateaus are elevated above sea level).

The grey area is known as the Basin and Range region – The middle part of it is a lower elevation flat(ish) zone flanked by mountains (the ranges) on either side – the Rockies to the east and the Cascades and Sierra Nevada ranges to the West.

The orange area in the American southwest is desert.



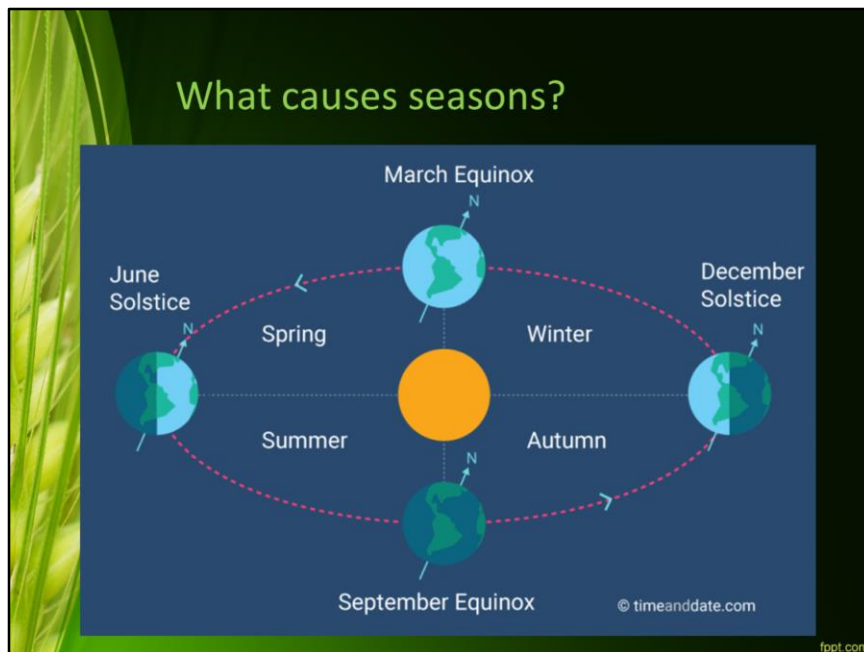
Climate regions are zones of the earth's surface with distinctive/predictable patterns of temperature and precipitation. Certain crops (plants) grow best in

The **Tropical Climate zone** is characterized by warm temperatures year-round and abundant (high amounts) of precipitation. The tropical regions are centered around the equator and are shown in red on the map above: South America, Central Africa, and Southeast Asia.

The **Desert Climate Zones**, shown in orange, are arid (dry) – they receive very little rainfall. The major deserts of the world are located on either side of the tropical zone.

The **Polar/Tundra zones**, shown in dark-blue, are cold year-round. The ground is typically frozen or covered with ice/snow for a majority of the year. This zone is not suitable for agriculture.

The **Temperate zone**, (also referred to as the “mid-latitudes”), shown in green, is the region of the earth located between the polar zone (frigid) and the “torrid” zone (hot – tropics and deserts, and the long-summer, short winter sub-tropical zones). This is the region of the world that has four distinctive seasons (winter, spring, summer, autumn). The length of time between the last frost of the spring (usually around March) and the first frost of the fall/winter (Usually early October) is known as the “growing season”.



Seasons are caused by the earth's tilt (23.5° off the vertical). The earth always points the same direction in space as it moves around the sun, which causes different parts of the earth's surface to have more direct or less direct sunlight at different times of the year.

Summer months: When the northern hemisphere is pointed toward the sun, days are longer (more hours of sunlight) than nighttime. The sun's rays are also passing through the earth's atmosphere at a more direct angle. This means the earth's surface in the northern hemisphere receives more energy from the sun during the summer months.

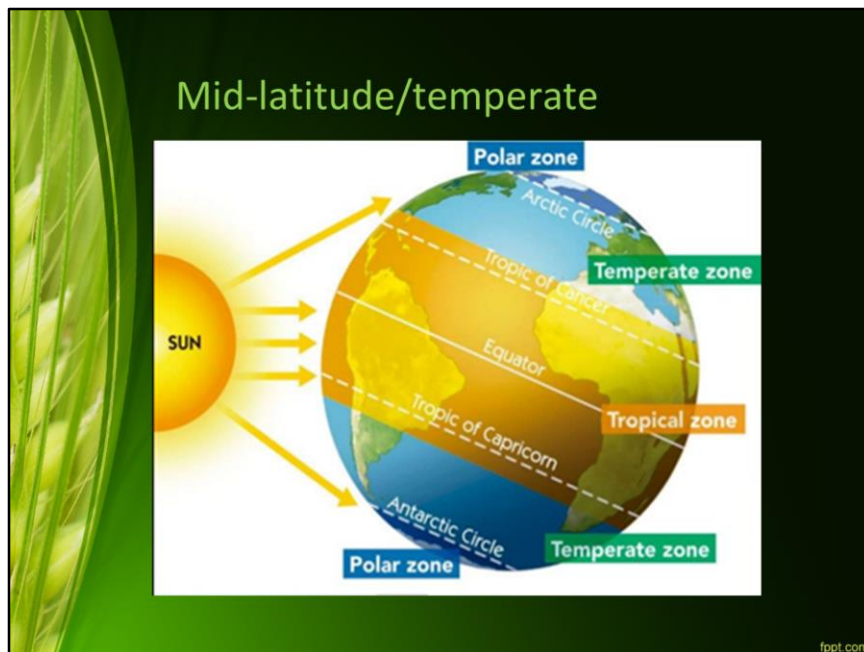
Winter months: When the northern hemisphere is tilted away from the sun, days are shorter (more hours of darkness than daylight) and the sun's rays are passing through the earth's atmosphere at a much lower angle – this means the sun's energy is spread out over a larger area and much of it is filtered by the atmosphere, meaning the earth's surface in a given location receives less heat energy from the sun during the winter months.

The Southern hemisphere is in winter (tilted away from the sun) while the northern

hemisphere is in summer.

The Southern hemisphere is in summer (tilted toward the sun) while the northern hemisphere is in winter.

The polar areas, despite having 24 hours of sunlight during the summer months are always cold because the sun's rays are filtered by the atmosphere and coming in at a low angle due to the spherical nature of the planet. This is why they have ice caps. They also have 24 hours of nighttime during the winter months (the sun never rises above the horizon).



Agriculture began in the temperate/mid-latitude climates. This climate zone had fertile soil, moderate amounts of rainfall at the right times of year (too much rain/flooding or too little rain at the wrong time of the growing season can be disastrous to that year's crop). Farming began in these climate regions and spread to other areas with similar climates. The techniques used by farmers were suitable to the soils and landscapes found in this region.

Cropping (growing plant-based food) is the primary use of land in the temperate climates. Large areas of trees have been cleared to create farmland in both Europe and North America. Grasslands are well suited for growing cereal grains.

Throughout history, until recently when the theory of environmental determinism was rejected in favor of the environmental Possibilism theory, early geographers (those of the colonial and early industrial ages) wrote of how the temperate zone was more suited to human development and therefore the people that lived in the temperate zone were more civilized and more technologically advanced than those people who lived in the torrid (hot) zone or the frigid (polar) zones. This partiality, unfortunately, was used to justify European imperialism in Asia and Africa as well as horrible atrocities of human history such as forced assimilation of natives and the institution of slavery.

Mediterranean Climates

- Temperate Zone
- Western side of continents



The map above shows the location of the Mediterranean climate zones of the world – there are 5 of them – the Western coast of the US, the Western coast of Chile that falls outside of the tropics, the southern coast of South Africa, South and Southwest Australia, and its namesake the Mediterranean basin (the area around the Mediterranean Sea).

This climate is characterized by warm dry summers and mild (not freezing) wet winters. The physical geography that lends itself to Mediterranean agriculture is distinguished by inconsistent rainfall, mild temperatures, uneven topography, and nearness to large water bodies.

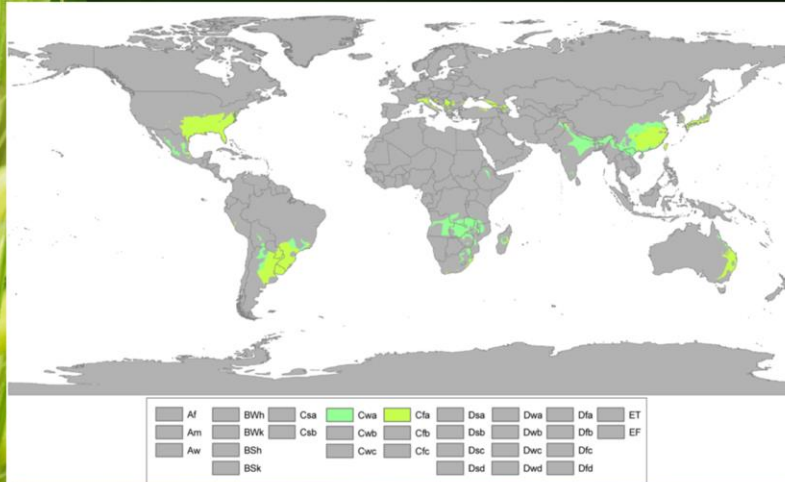
Mediterranean Crops (those that thrive in this climate zone) include citrus fruits, olives, and figs, which come almost exclusively from Mediterranean climates. **Cereal grains** (grasses with edible seeds) are extremely important to agriculture in this region, primarily wheat and barley. Other fruits and vegetables also thrive in the region, including eggplant, chili peppers, potatoes, tomatoes, broccoli, cabbage, zucchini, cucumber and pulses like chickpeas, peas, and lentils.

Mediterranean Animal husbandry (animal husbandry means breeding and caring for farm animals) include goats, sheep, and cattle. In mountainous areas, farmers practice **transhumance** where farmers move the animals up the mountain to green pastures in the

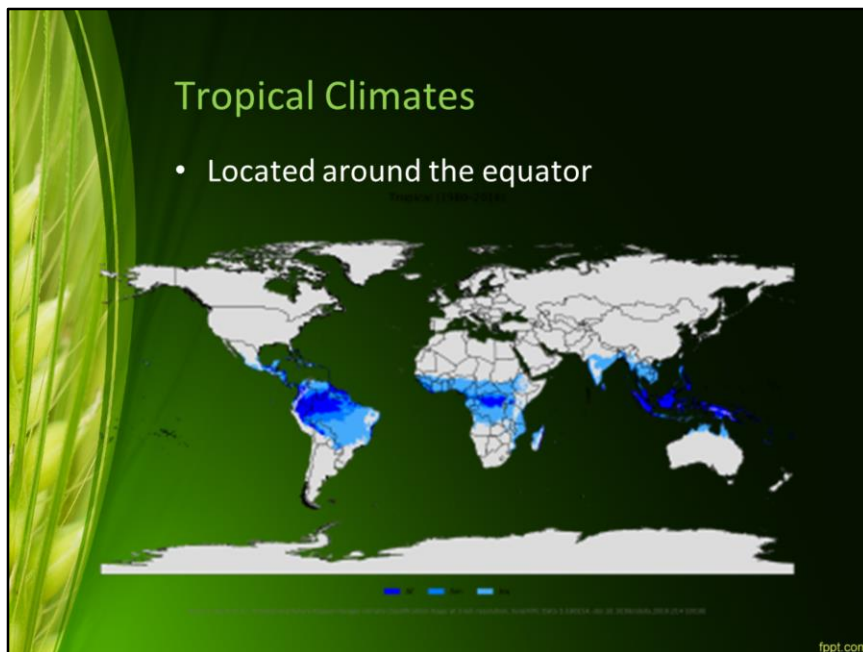
summer and returning them to the lower elevation pastures in the valleys in winter.

In the Mediterranean Basin, commercial farming (farming for profit) and subsistence farming (farming for ones own consumption) happen side-by-side.

Sub-tropical



Sub-tropical climates are located just outside of the tropical zone. These climates have hot summers and mild winters, with rain throughout the year. Certain crops grow very well in this climate region, that has a long growing season (often more than 200 days between frosts) – particularly cotton, tobacco, citrus fruits, avocado, nuts (peanuts, pistachios, pecans, almonds, and others), and a large variety of squashes, cauliflower, broccoli, and other vegetables that grow well in warm soil.



Tropical climates have an average temperature above 64° year round – this means it does not freeze. Tropical climates centered around the equator only have one season – there is not a seasonal shift as the sun's ray's are almost always directly overhead. Just outside of the equatorial zone, there are two seasons – the wet season and the dry season.

Tropical agriculture has different needs than agriculture in other regions. Traditional methods of agriculture developed in the temperate/mid-latitude and sub-tropical regions are not well suited to the tropical zone. Here, plants can be grown in the tropical season year round – this makes it much more suitable to propagate plants from cuttings (growing a new plant from a cutting (leaf, stalk, etc) instead of from a seed. Seeds are necessary to delay planting in times of freezing (winter) or drought, but those conditions are not an issue in the tropics.

Unfortunately, propagating plants in this fashion makes them all genetically identical, and therefore more vulnerable to disease. For example, the Gros Michel banana was the most popular variety of banana world-wide. It was wiped out by a fungus known as "Panama Disease." Today, the most common variety of banana, and the one you are most familiar with, is the Cavendish banana, which is also at threat of a new fungus spreading through southeast Asia.

Traditional methods are especially damaging to the environment of the tropics, resulting in deforestation, soil degradation (loss of soil fertility), erosion, and pollution of waterways from

fertilizers and chemicals. The region is also highly susceptible to climate change.

While growing food for local consumption (subsistence agriculture) is the primary goal of tropical agriculture, cash crops (crops grown for profit, typically for export – which is a legacy of European colonialism).

Cash crops typically grown in the tropics: bananas, pineapple, coconut, cocoa (used to make chocolate), coffee, and sugarcane.



Commercial farming is a type of farming in which crops are grown for profit/sale only. It is a modernized method of farming that is undertaken on a large scale in order to grow a surplus of food for sale. The more food a farmer can grow, the more can be sold, and the more money he or she can make. Commercial farmers often specialize in one or two types of crops in order to maximize their profits. In this type of farming, large plots of land, labor (people), and machines are used (image above left).

Subsistence agriculture occurs when farmers grow food crops to feed themselves and their families. In subsistence agriculture, farm output is targeted to survival and is mostly for local requirements with little or no surplus for trade or sale. The farmers have smaller plots of land, rely on human and animal labor, and grow a wide variety of crops to maximize the range of nutrients available to them throughout the year (image below right).



Intensive agriculture, also known as intensive farming, is a type of agriculture (both of crop plants and of animals) with higher levels of input and output per cubic unit of agricultural land area. The inputs (labor, machinery, fertilizers, pesticides, specific varieties of crops, etc) are high in intensive farming. The farmers invest a lot in terms of inputs in order to maximize their output (referred to as crop yields). With intensive farming, the amount of food produced per unit of land (i.e., per acre or hectare) is high. Huge amounts of food (including plants and animals) can be produced off a relatively small amount of land.

Intensive agriculture can be either commercial (grown for profit, typically with machinery, chemical fertilizer and pesticides) or subsistence (grown for own consumption and dependent on more human labor and natural methods of fertilizer and preventing insect damage).



Market gardening is the term used to describe the commercial production of vegetables, fruits, flowers and other plants on a scale larger than home gardening, but smaller scale than huge corporate farms. Market gardens typically grow food for nearby cities as the produce is **perishable*** in its raw form and needs to be sold and consumed quickly. Market gardens are intensive agriculture as the farmer must invest in quality plants (often selected varieties that are suitable for their climate zone), fertilize the soil regularly, and irrigate. Greenhouses are also used to extend the growing season, sometimes year-round. The yield is very high for a small plot of land.

*perishable: (especially of food) likely to decay or go bad quickly. Milk, fruits, vegetables, flowers, etc. that have not been processed or preserved (by jarring or freezing) are perishable.

Images: Left, tomatoes growing in a greenhouse. Right, a market garden with a variety of crops.



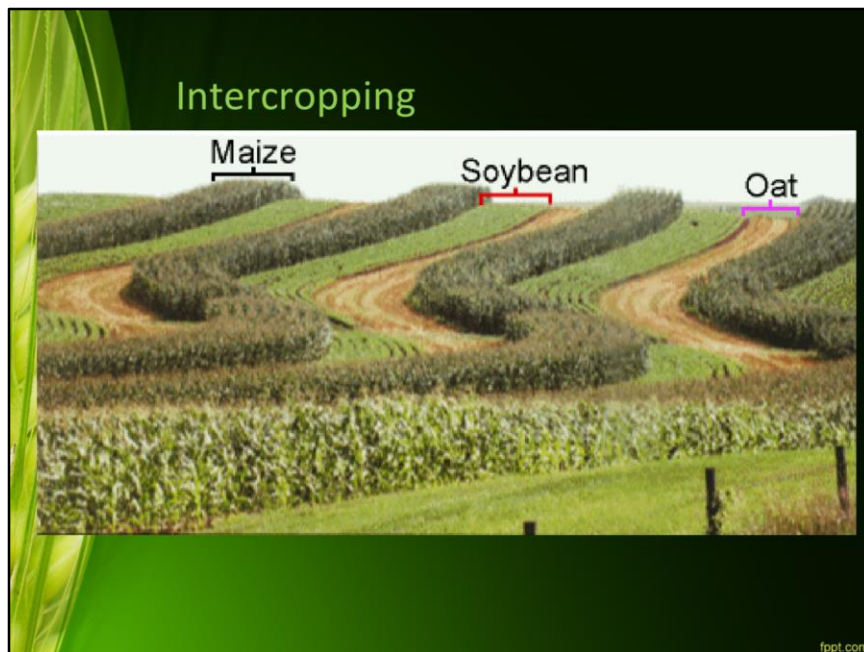
Plantation agriculture is an intensive form of commercial farming where crops are grown for profit. Large amounts of capital investment (money) are needed to start a plantation. Large land areas are needed, as well as machinery or an inexpensive labor force (low wage workers) in order to maximize profits. Plantation owners typically focus on growing one or two crops in order to minimize the costs and maximize their profits through economies of scale. Plantations typically range from 500 to 1,000+ acres of land (there are 640 acres in a square mile for comparison). Countries that have plantation agriculture usually experience tropical climate with high annual temperatures and receive high annual rainfall, or a sub-tropical climate where crops are suitable to the plantation method. Large amounts of fertilizers and pesticides are often used in plantation farming in order to increase the yield as much as possible. Typical plantation crops: cotton, coffee, cocoa, rubber, palm, sugarcane, tea, tobacco, banana, pineapple, etc. Some plantations you may have heard of: Chiquita Banana, Dole.

Plantation farming dates back to the Roman Empire, where wealthy families owned land and used slaves or low wage workers to do the labor. The plantation system diffused across Europe and eventually to the Americas with the European colonists. Thus, the first two centuries of European settlement in the southern part of North America firmly established the definition of a plantation: a very large farm that used slave labor to produce a commodity for export (cotton and tobacco in North America).

Images: A worker picks coffee off a plant on a plantation in Indonesia (left), a man tours a banana plantation (right).

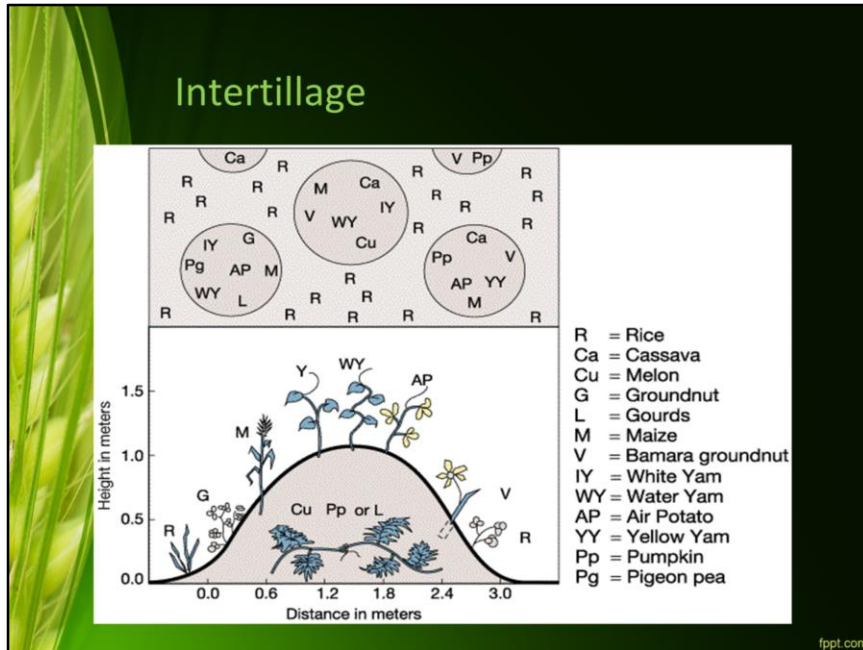


A typical **mixed crop and livestock farm** devotes nearly all land area to growing crops but derives more than three-fourths of its income from the sale of animal products, such as beef, milk, and eggs. The crops are often used to feed the animals (hay, corn, etc.). This is a form of intensive agriculture (it is different than ranching).



Intercropping is a multiple cropping practice involving growing two or more crops in close proximity. The most common goal of intercropping is to produce a greater yield on a given piece of land by making use of resources or ecological processes that would otherwise not be utilized by a single crop. This practice is commonly used in both commercial and subsistence agriculture.

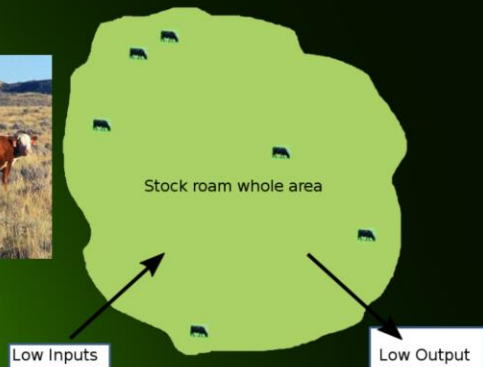
Intertillage



Intertillage is a subsistence agricultural practice, commonly used in slash & burn agriculture, which spreads out production over the farming season by planting different crops in the field. More specifically, it is the practice of planting crops along the same row or mound. The image above illustrates planting a variety of plants on each mound, with rice (which requires the field to be flooded) taking the ideal location at the base of the mounds, but a variety of other plants are planted at varying levels of the mounds. Intertillage requires substantial human labor as machinery would not work for such a system. This practice is common in Southeast India, Southeast Asia, and East India and Southeast China for their rice crops.

Extensive Farming

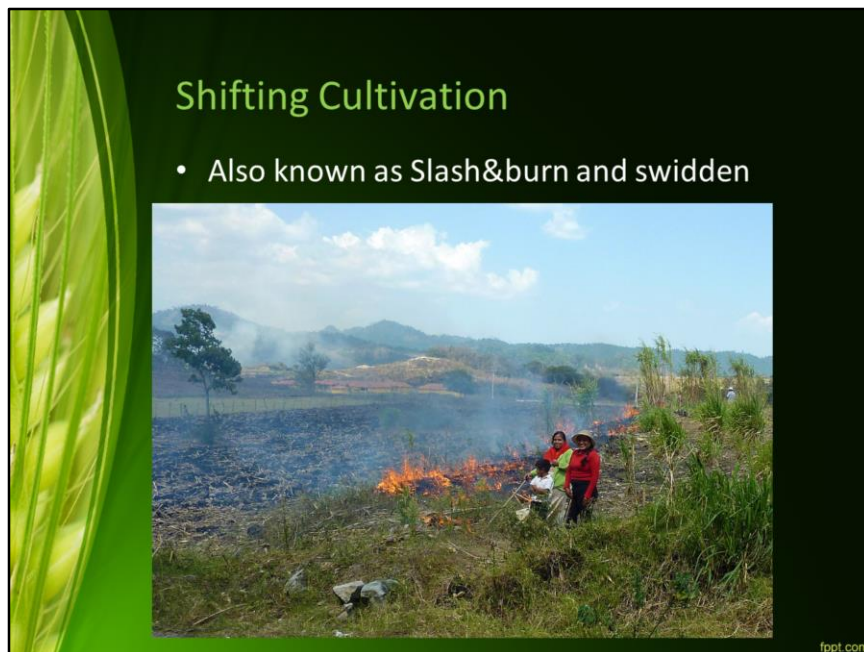
- Low \$ inputs, low yield
- Large amounts of land required



fppt.com

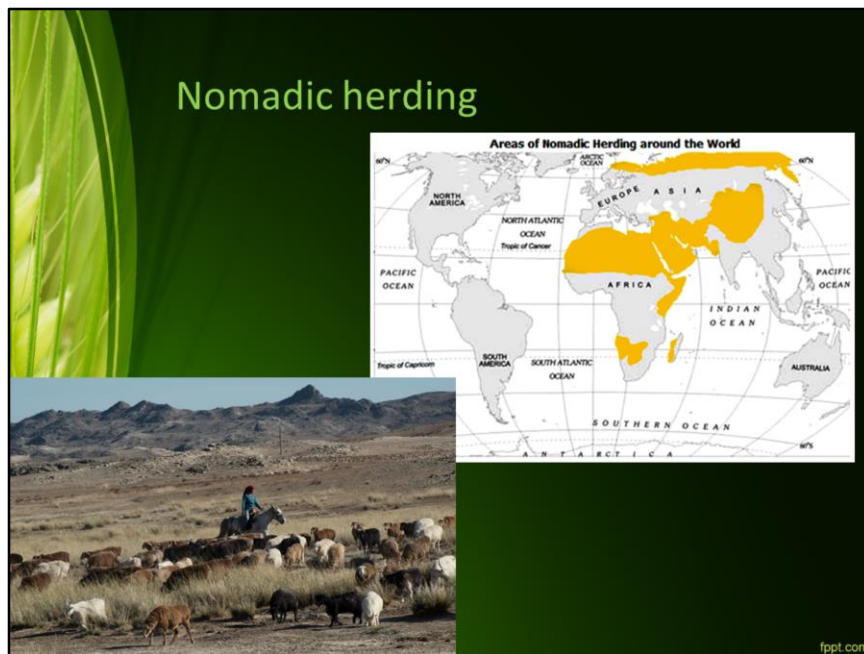
Extensive farming is an agricultural production system that uses small inputs of labor, fertilizers, and capital, relative to the land area being farmed.

Examples of extensive farming include ranching (grazing animals), nomadic herding, and shifting cultivation.



Shifting Cultivation is a form of agriculture, used especially in tropical Africa, South America, and Southeast Asia, in which an area of ground is cleared of vegetation and cultivated for a few years and then abandoned for a new area until its fertility has been naturally restored.

A large amount of land is needed because the farmer clears a few acres at a time for planting. The trees and other vegetation are cut down and then burned, with the ashes providing nutrients to the soil. The soil is only fertile for a few years before it loses its nutrients to leaching and needs to be replenished. The farmer then moves to a new plot and allows the previously farmed land to regrow the natural vegetation. In order for this practice to be sustainable, the farmer needs to let the land lie fallow (unused) for a number of years – up to 20. That means the farmer needs to have a lot of land, though only a small portion of it is used at any one time and the yield from the total acreage is low. Unfortunately, population and development pressures have pushed subsistence farmers who practice shifting cultivation onto smaller and smaller amounts of land. Instead of returning to a plot after 20 years, they are forced to return in just 2-5 years, which means the soil has not had time to replenish the nutrients. As a result, the soil loses its fertility even faster, or the farmers are pushed up hillsides where there is a greater risk of soil erosion. Deforestation is also a major effect (deforestation is a major effect any time trees are cleared to create more farmland).



Nomadic herding, sometime referred to as **pastoral nomadism**, is the practice of caring for roaming groups of livestock over a large area. Herding developed about 10,000 years ago, as prehistoric hunters domesticated wild animals such as sheep and goats. Hunters learned that by controlling animals they once pursued, they could have reliable sources of meat, milk and milk products, and hides for tents and clothing. Nomadic herders roam in small tribal or extended family groups and have no home base. Nomads live in arid and semiarid parts of Africa, Asia, and Europe, and in the tundra regions of Asia and Europe.

The yellow areas on the map above are the parts of the world where this extensive form of agriculture takes place. Notice these areas are deserts or tundra, and are otherwise not well-suited for crop agriculture. Nomadic herding is considered subsistence agriculture, even though the herders often trade meat, milk, and furs for other products, it is just enough for them to survive on, not to make a profit.

Nomadic herders also face the pressures of population growth, also as nomadic tribes have no true “home base”, many states try to limit their movements across international borders. As a result, the tribes are being pushed into smaller and smaller areas (though they still occupy and use large amounts of land across the continents of Africa and Asia).

Ranching



fppt.com

A ranch is an area of land, including various structures, given primarily to the practice of ranching, the practice of raising grazing livestock such as cattle or sheep for meat or wool. Unlike nomadic herding, ranch land and pastures are set by property lines and often fenced off from neighboring pastures. Ranching is a commercial farming activity.

Ranch farming originated in the introduction of European livestock-farming techniques onto the vast open grasslands of the New World (the Americas). The invention of barbed wire (is a type of steel fencing wire constructed with sharp edges or points arranged at intervals along the strands) in the mid 1800s made it inexpensive for ranchers in the US to fence off huge parcels of land (1,000s of acres at times). Barbed wire played an important role in the protection of range rights in the Western U.S. Although some ranchers put notices in newspapers claiming land areas, livestock continued to cross range boundaries. Fences of smooth wire did not hold livestock well, wooden fences were cost-prohibitive, and hedges were difficult to grow and maintain. Barbed wires introduction in the west in the 1870s dramatically reduced the cost of enclosing land the mostly flat grassland plains of the central portion of the US. There's a saying that the land is so flat, there's nothing between Texas and the North Pole except a barbed wire fence.

Classification of Rural Settlements



fppt.com

Where people farm, they need a place to live. However, these settlements differ greatly than urban settlements in that there is often little more than a few houses scattered across the horizon or clustered into a group surrounded by farmland. In this next section of the notes, we'll look at notable patterns of rural settlements and then survey methods associated with each type of settlement.

Clustered Settlements



A **clustered rural settlement** is an agricultural- based community in which a number of families. live in close proximity to each other, with fields. surrounding the collection of houses and farm.

Clustered settlements occur in rural agricultural communities all over the world. The photo top left is in Wales (UK), the photo on the right is in Kenya, and the picture on the bottom is in Romania.

Linear Settlements

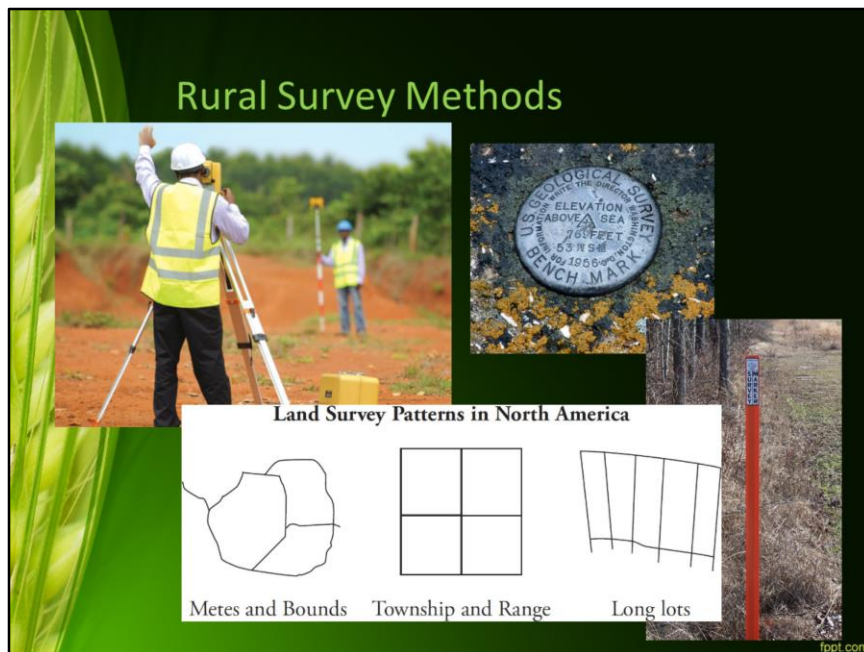


Linear settlements are a group of buildings that are formed in a long line. Many follow a transport route, such as a road, river, or canal.



A **dispersed settlement**, also known as a scattered settlement, is one of the main types of settlement patterns used by landscape historians to classify rural settlements found in England, the US, and other parts of the world. Typically, there are a number of separate farmsteads scattered throughout the area.

In the United States, dispersed settlements were common in the colonies, but became even more prevalent as the US expanded westward. This new land was surveyed (measured) in a way that would allow the US government to space people out across the land in what has become a very distinctive landscape pattern (see Township and Range system).



A **land survey**, or simply **survey**, is the scientific process of measuring the dimensions of a particular area of the earth's surface, including its horizontal distances, directions, angles, and elevations. Artificial structures, such as a road or building, may also be noted on a survey. These measurements are noted along with a description of the property for mapping and to record property ownership.

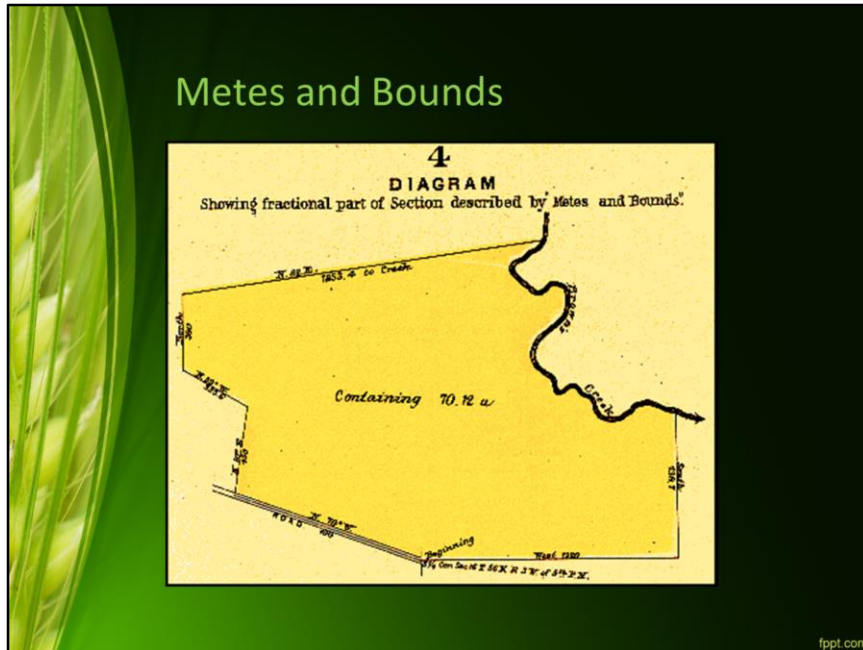
This part is background information – you should read it, but you won't be tested on it.

The process of surveying is actually pretty simple. A measuring wheel or a long measuring tape is used to measure the distance between the person with the tripod, who uses an instrument called a theodolite to take measurements off the pole the other person is holding (called a stadia rod). The measurements tell the surveyor the difference in elevation between the location of the two instruments. The measurements are recorded, and the two people reposition to take new measurements until the entire area has been measured. Can you imagine surveying the entire expanse of the Louisiana purchase or the rest of the North American continent with chains or string and some measurement tools to map it before GIS was invented? This is how it was done, on foot and taking measurements bit by bit. As an area is surveyed, markers called "benchmarks" are placed that state the elevation and some other information. The results were not always very accurate, which led to some disputes over boundaries between the US states and between the US and Canada. The US and Canadian border is an interesting story as it is set along the 49th parallel, but when it was demarcated it was anything other than a straight line. For an amusing video on the US

Canadian border, watch this video: <https://youtu.be/qMkYIIA7mgw>.

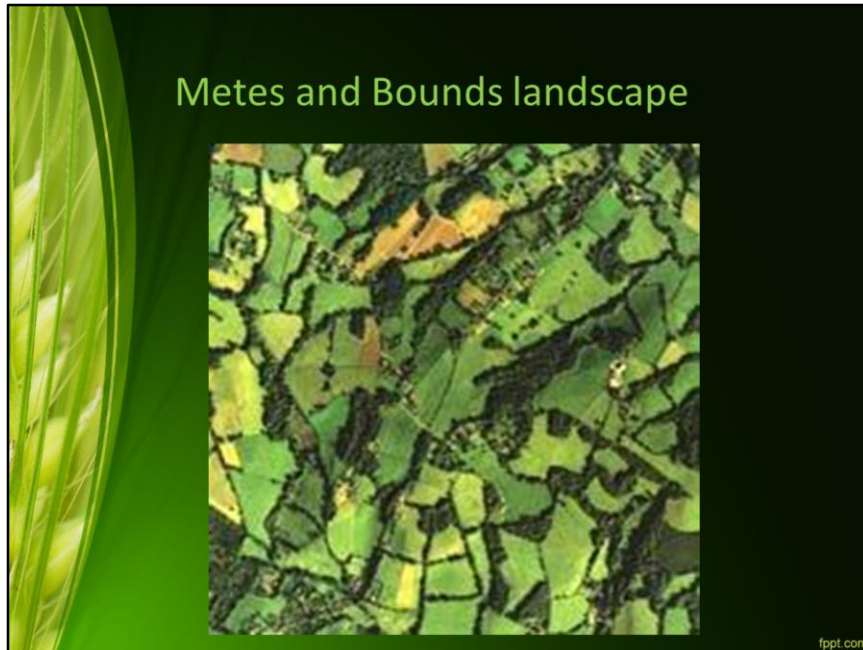
There are different ways of recording the surveyed data, which are known as “survey systems”. There are many different survey systems in use today, but you need to be familiar with three of them: the metes and bounds system, the long lot system, and the township and range system (more info on each to follow).

Metes and Bounds



The **metes and bounds** system was the old English surveying system that the colonists brought over to the Americas with them (tie back to prior unit: relocation diffusion).

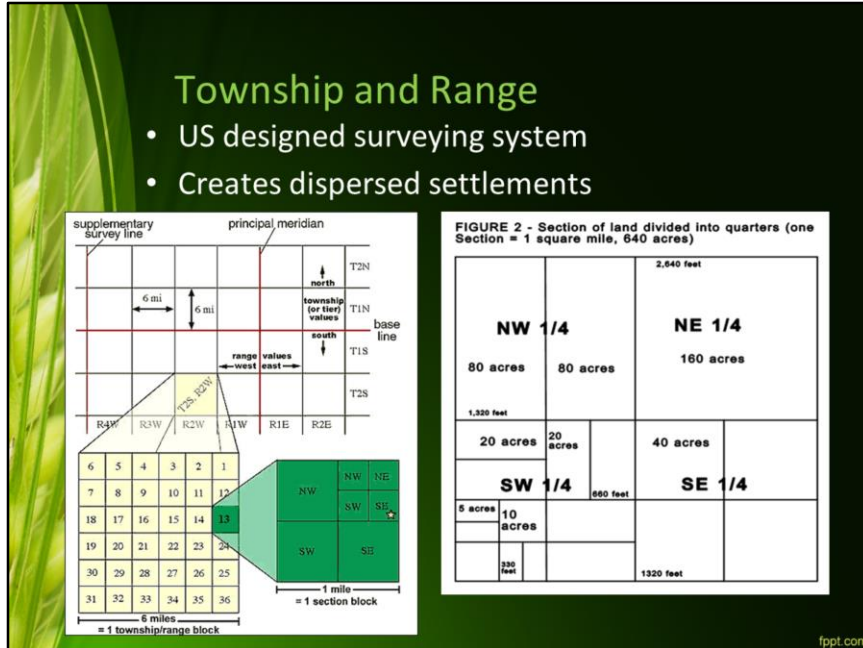
The **metes and bounds** surveying system uses landmarks including rivers, roads, stakes, or other such natural or manmade markers. **Metes and bounds** descriptions are typically used whenever land survey areas are irregularly sized and/or shaped, so that the boundaries of the property can be more clearly defined. As a result, properties or political boundaries surveyed using this system (typically after it was settled) have irregular shapes and tend to be smaller.



The metes and bounds systems creates irregularly shaped properties, as shown on the map above. The darker green areas dividing the properties were created when brush and trees grew up around fences.

Township and Range

- US designed surveying system
- Creates dispersed settlements



The **Township and Range System**, also known as the Public Land Survey System, is the surveying method developed and used in the United States to plat, or divide, real property for sale and settling. It is based on latitude and longitude “baselines”. TOWNSHIP measures the distance north or south from the baseline, which is a designated parallel (line of latitude). A township usually measures 6 miles in size. RANGE measures east or west of the principle meridian (line of longitude) which is a designated meridian. Each section is approximately 36 square miles, which is then subdivided into 36 sections of 640 acres each. These sections are further sub-divided as shown in the illustration on the right above.

It sounds confusing – but what you really need to know is that the Township and Range system was a system of surveying the sparsely population (or uninhabited) land and mapping it in a nice, neat, and orderly rectangular system, then dividing each of the rectangles into smaller plots and granting/selling them to people who wanted to move out into the frontier and start a farm. It was a way for the US to spread the population out over the Great Plains. Roads were built on the boundary between each, leaving a distinguishable pattern of demarcated boundaries on the landscape of the “flyover states*”.

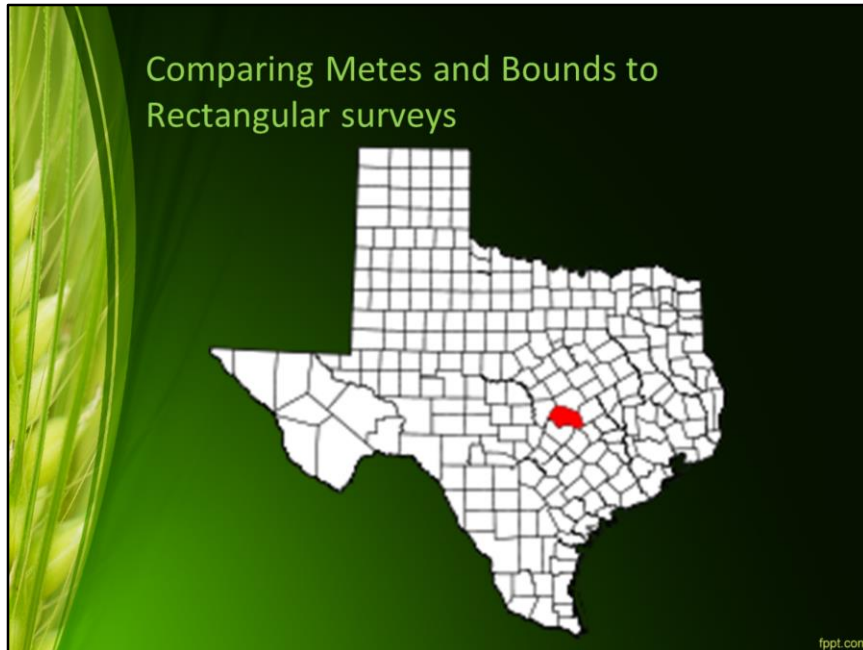
A **flyover state** is a mildly derogatory term referring to the central states of the US that are often passed over by intercoastal/transcontinental flights, rather than being a popular destination.

Township and Range System



fppt.com

The image above is an aerial photograph taken of land surveyed using the township and range system. Notice how the roads are even spaced apart and intersect at right angles. The houses/farmsteads are spread out fairly evenly in a dispersed pattern.



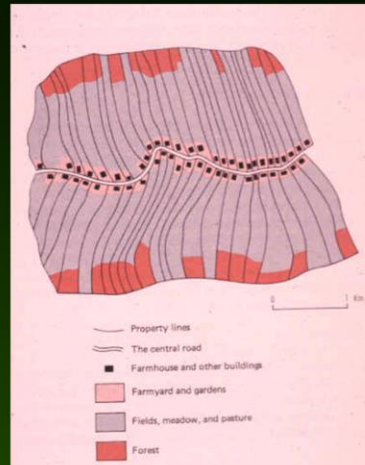
To see the difference between the metes and bounds survey system and a rectangular (rectilinear) surveying systems, just look at the counties of Texas.

The counties in the eastern part of the state – the first part of the state that was settled by Americans – was surveyed using the metes and bounds system. These counties are small and have irregular shapes, including Williamson county (the county we live in – shown on the map in red).

The counties in the panhandle and western part of the state were surveyed prior to being settled (antecedent boundaries) and are based on a rectangular survey system. They are more even in size (except for the counties in the desert and mountainous part of Texas, which are huge and sparsely populated).

Long Lots

- French survey method
- Creates linear settlement patterns



Areas in North America (Quebec and Louisiana) settled by the French used the French survey system known as the Long Lot System. Each property (lot) is long and narrow and has access to the river, canal, or road. This system gave farmers access to the transportation route while keeping the size of the parcels relatively equal. Each farmer could also live on their own property and still be close to neighbors (linear settlements), compared to clustered settlements where the farmers lived close to their neighbors in the village, but disconnected from their farm, or in dispersed settlements where the farmers lived on their farm but were far apart from their neighbors.

Of course, this system had challenges – rivers are rarely straight and change course over time, which would affect the property and cause possible future legal issues for the property owners. Additionally, the properties on the inside of a river bend would be cut short. Another issue was the French inheritance system in which the property of the father was evenly distributed among his children upon his death – the lots were divided long-ways, which was usually not a huge issue if there were only two kids, but French Canadians traditionally had large families, so if the lot was divided evenly 6 or 7 ways, the strips became too narrow to be functional.

Long Lot System



This is a screen shot of a satellite image taken of Saint-Francois-de-l'Île-d'Orleans, in Quebec.